

aeris_pcb

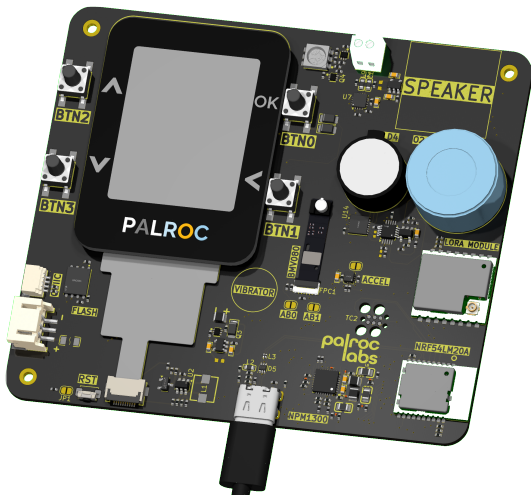
Variant: PRELIMINARY

2026-07-11

Rev + (Unreleased)

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TOP VIEW



BOTTOM VIEW



NOTES

Comment

Not fitted components are marked as **X**

DRAFT - Very early stage of schematic, ignore details.
PRELIMINARY - Close to final schematic.
CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.
RELEASED - A board with this schematic has been sent to production.

Date: 11-Jul-2026

DESIGN CONSIDERATIONS

DESIGN NOTE:

Example text for informational design notes.

DESIGN NOTE:

Example text for debug notes.

DESIGN NOTE:

Example text for cautionary design notes.

DESIGN NOTE:

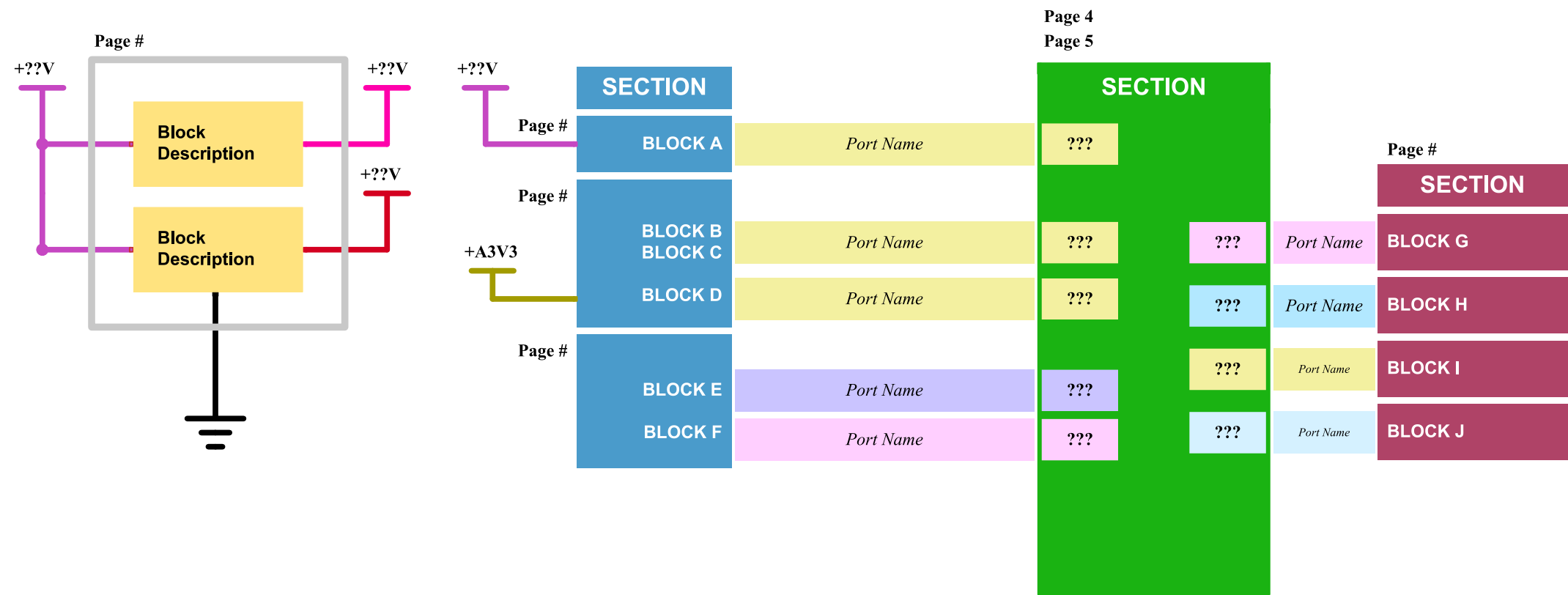
Example text for critical design notes.

LAYOUT NOTE:

Example text for critical layout guidelines.

	Comments:	Company: Palroc Labs	Variant: PRELIMINARY	Git Hash: 4dfdc44
	Sheet Title:	Board Name: aeris_pcb	Project Name: Palroc One	
	Sheet Path: /	File Name: palroc_one_pcb.kicad_sch	Designer: Oriol Reverte	Date: 2025-01-12
			Revision: + (Unreleased)	
			Size: A3	Sheet: 1 of 12

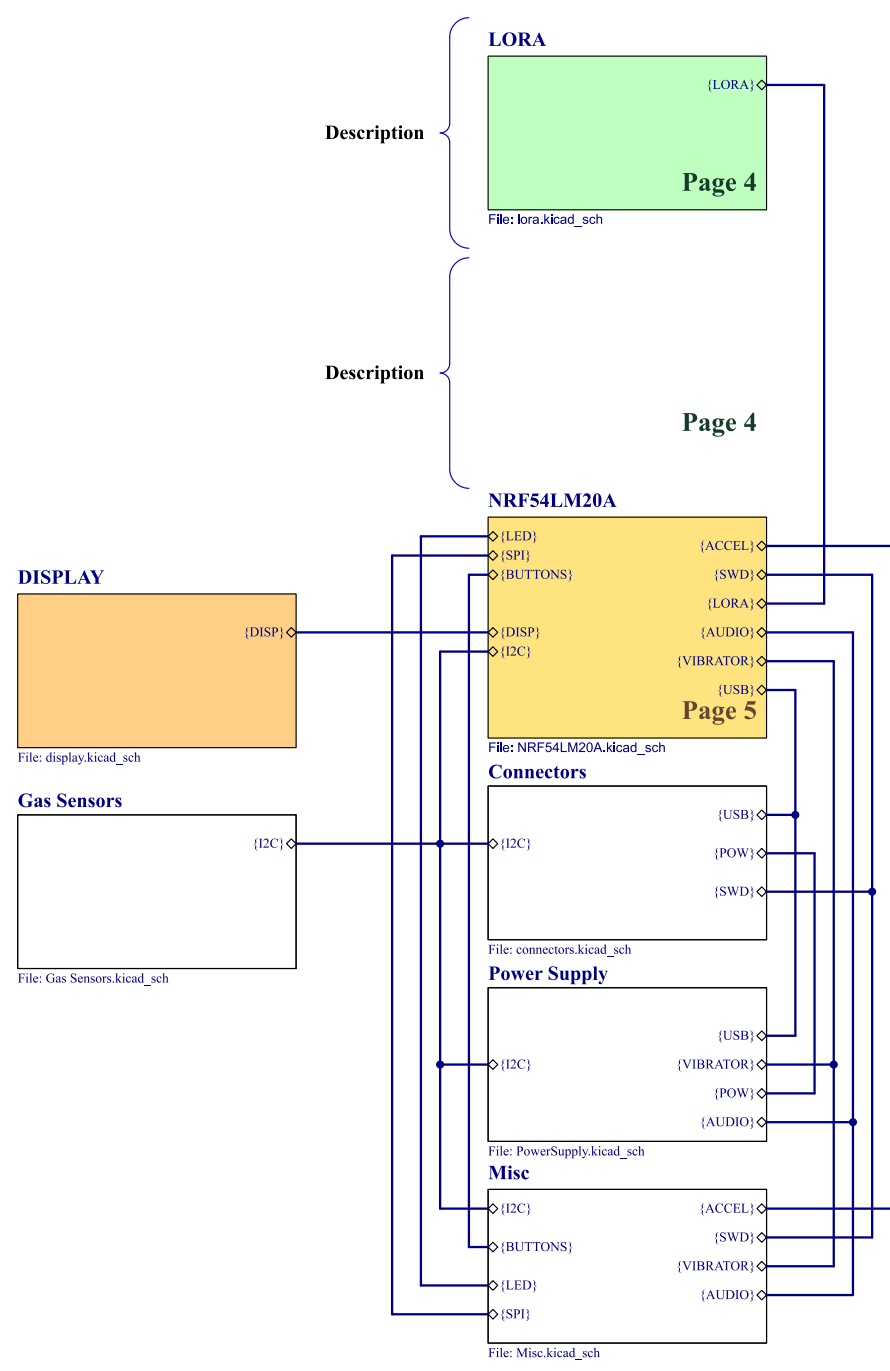
[2] Block Diagram



Target specifications:	
Input voltage:	?? - ?? V
Spec 2	??
Spec 3	??
Spec 4	??

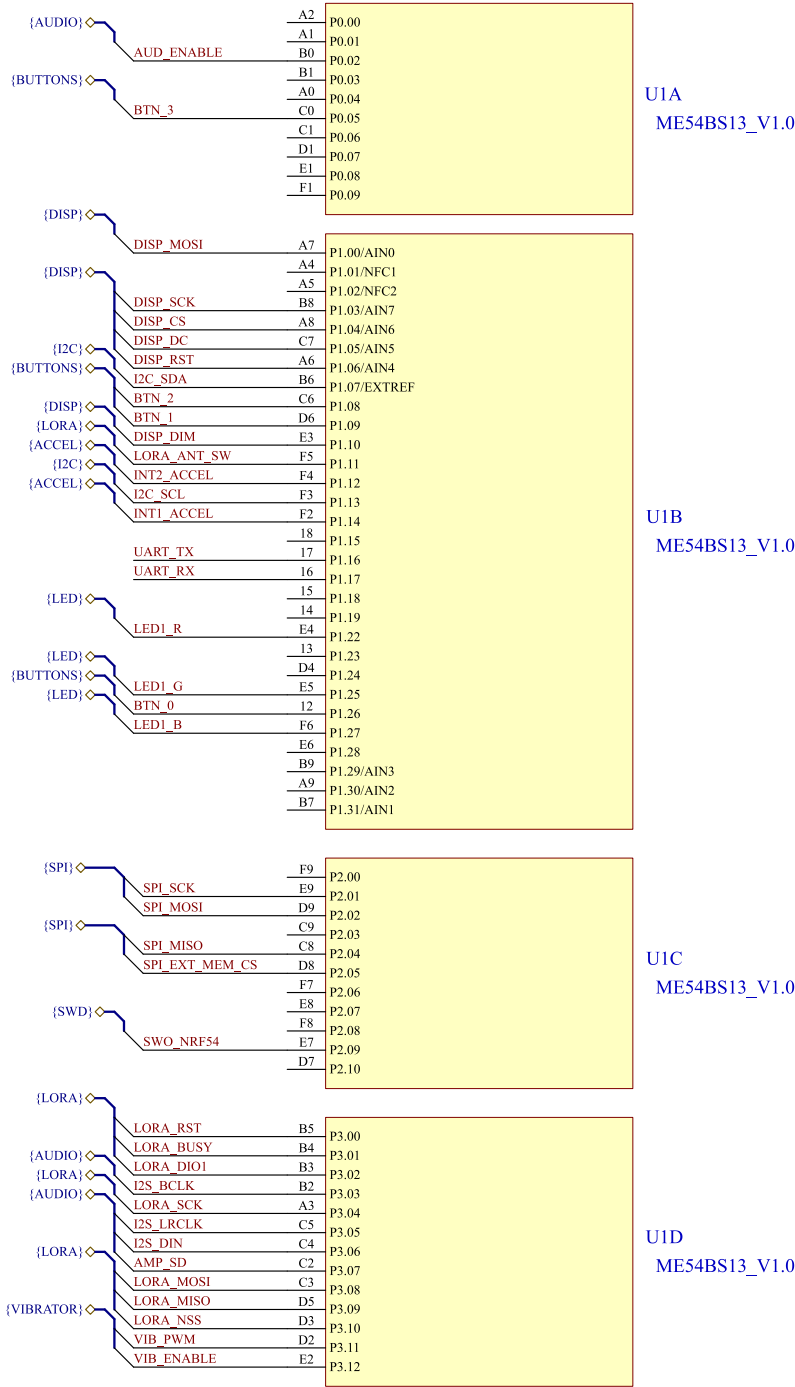
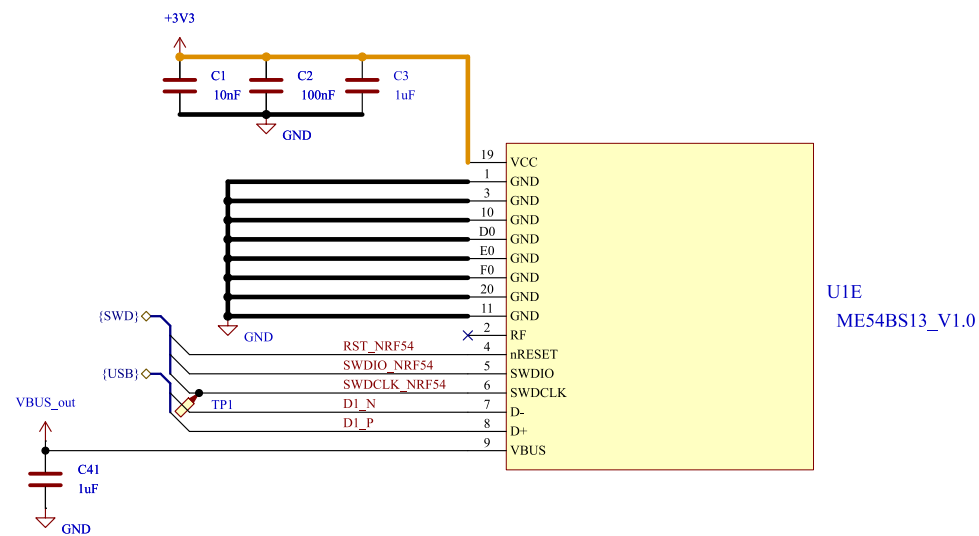
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	Sheet Title: Block Diagram	File Name: Block Diagram.kicad_sch	Designer: Oriol Reverte	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Block Diagram/			Reviewer:	Size: A3	Sheet: 2 of 12

[3] Project Architecture



	Comments:	Company: Palroc Labs		Variant: PRELIMINARY	Git Hash: 4dfdc44
		Board Name: aeris_pcb		Project Name: Palroc One	
	Sheet Title: Project Architecture	File Name: Project Architecture.kicad_sch	Designer: Oriol Reverte	Date: 2025-01-12	Revision: + (Unreleased)
	Sheet Path: /Project Architecture/		Reviewer:	Size: A3	Sheet: 3 of 12

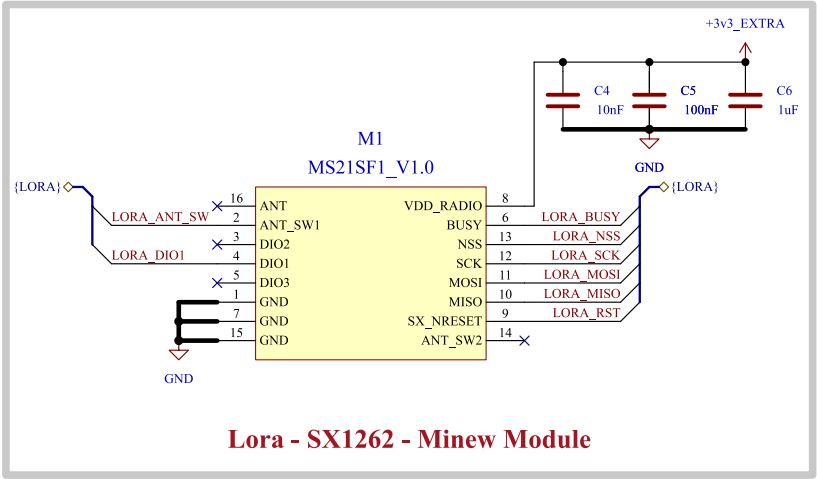
[5]
NRF54LM20A
Module



Block Description

Comments: AN5346 STM32G474 Datasheet p.81 J. Pieper ADC investigation	Company: Palroc Labs		Variant: PRELIMINARY		Git Hash: 4dfdc44	
	Board Name: aeris_pcb		Project Name: Palroc One			
Sheet Title: NRF54LM20A Module	File Name: NRF54LM20A.kicad_sch	Designer: Oriol Reverte	Date: 2025-01-12	Revision: + (Unreleased)		
Sheet Path: /Project Architecture/NRF54LM20A/		Reviewer:	Size: A3	Sheet: 5 of 12		

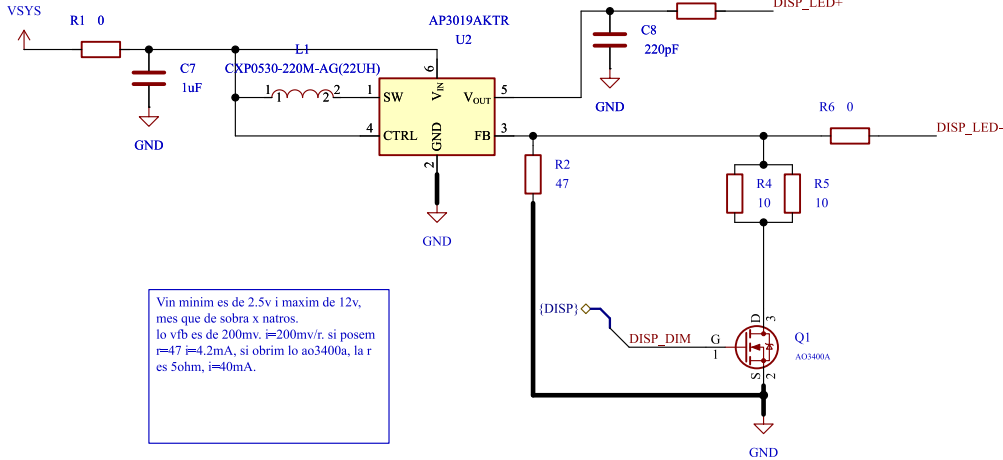
[8] Lora Module



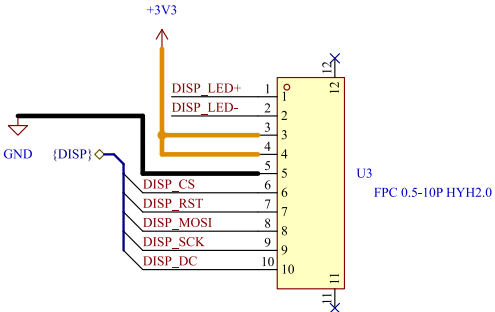
Lora - SX1262 - Minew Module

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	Sheet Title:	File Name:	Designer:	Date:	Revision:
	Lora Module	lora.kicad_sch	Oriol Reverte		
	Sheet Path:	Reviewer:		Size:	Sheet:
	/Project Architecture/LORA/			A4	8 of 12

[9] Display



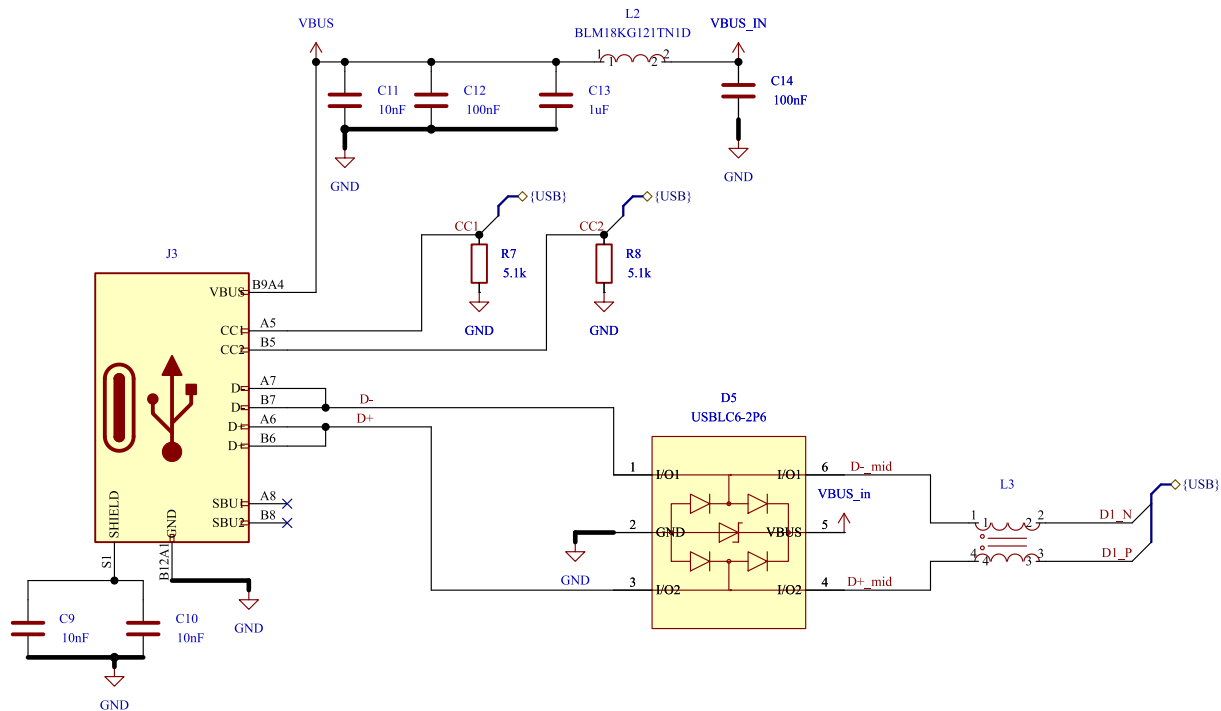
Display Power IC



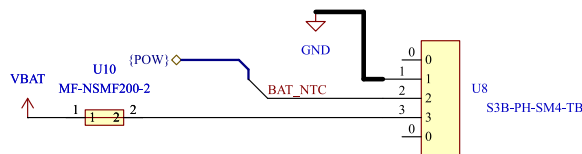
Display Connecto

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		Board Name: aeris_pcb		Project Name: Palroc One	
	Sheet Title: Display	File Name: display.kicad_sch	Designer: Oriol Reverte	Date:	Revision:
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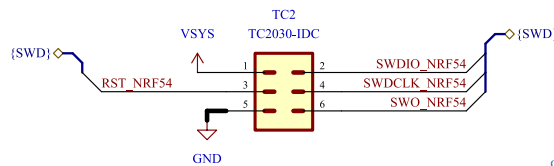
[10] Connectors



USB C Connector

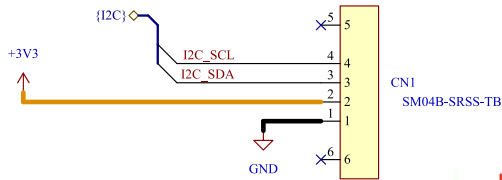


JST Battery Connector

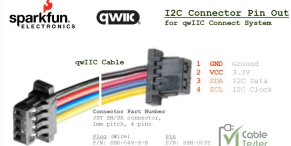


TC2030 Programming Pin

Connections:	
10-Pin Cortex Debug Connector	6-Pin TC2030 Footprint
1 VCC	1 VCC
3 GND	2 SWDIO / TMS
5 GND	3 nRESET
7 NC / RTCK	4 SWCLK / TCK
9 GNDdetect	5 GND (also connected to GNDdetect)
	6 SWO / TDO



JST Battery Connector



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	Board Name:		Project Name:		
	aeris_pcb		Palroc One		
Sheet Title:	File Name:		Designer:	Date:	Revision:
Connectors	connectors.kicad_sch		Oriol Reverte		
Sheet Path:	Reviewer:		Size:	Sheet:	
/Project Architecture/Connectors/			A3	10 of 12	

[11] Power Supply

No communication is needed between nPM1300 and the USB host. The CC lines can be used to detect the USB host max current capability, but even then it is not automatic, you can read the detected value in the register and SoC can set the VBUSINLIM accordingly through TWI. So you can just set the VBUS current limit to the desired value with TWI write when the 5V adapter is plugged. Note the default is 100mA, so that is the maximum you can draw from the 5V supply with the default setting. You can program it up to 1500mA.

nPM1300 settings:

- VOUT1: 2.7v
- VSET1: 300k
- VOUT2: 3.3V
- VSET2: 300k

Vout1 can only be 2.7V top trough VSET, but we can set trough sw the vout and it can be 3.3V

each of the bucks (vout1 and vout2) can provide up to 200mA, not more.
The vout1 will be 1v8 and will be used for the nrf7000, (which should have a maximum current consumption of 64mA in pasive mode) in active mode it consumes around 250mA on peak). We need to take that into account to stick with pasive.

NOTE: C55 and C56 should be placed close to PVSS1 and PVSS2 pads respectively.

VSYS: 2500mA max
Connected to:
-nr9151

Initially its 2v7. But it will be changed trough SW to 3v3

VOUT1:
120mA Max current from Lora
45mA from BMV080 Analog
22.4mA from BMV080 Digital
Total: 190mA

Buck 2: 200mA max
Connected to:
-IMU
-Flash
-GPIOs
-nr54lm20

A maximum of 50mA can be drawn x LDO

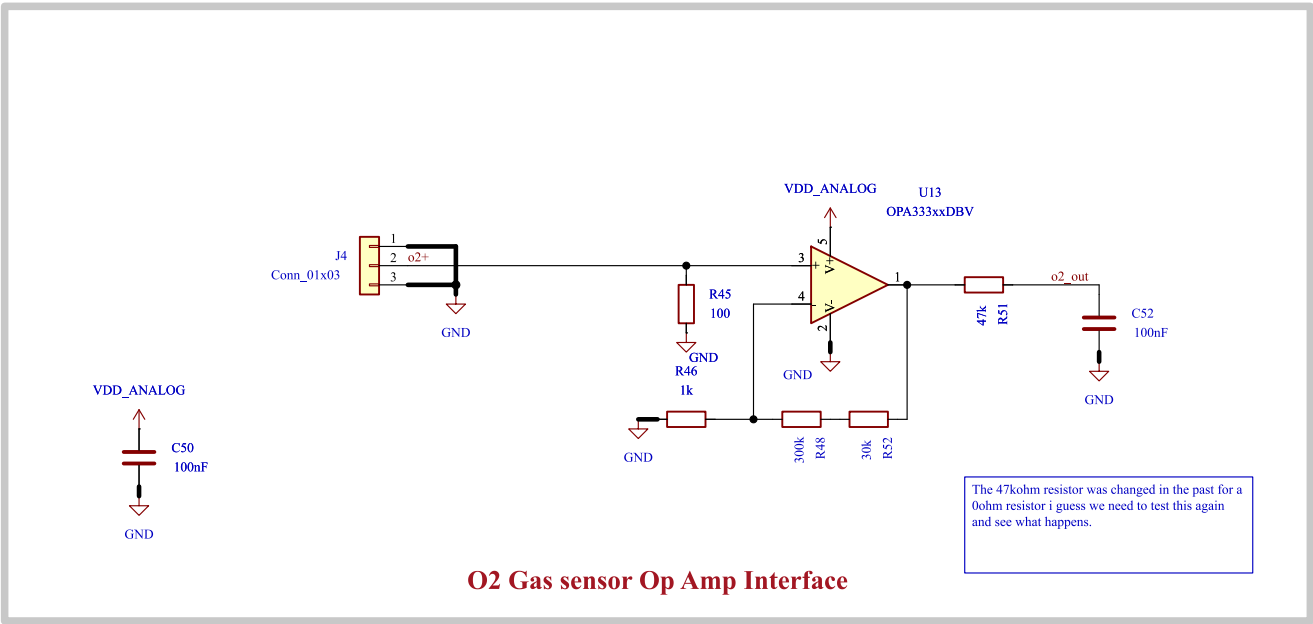
NPM1300

Vibrator power switch

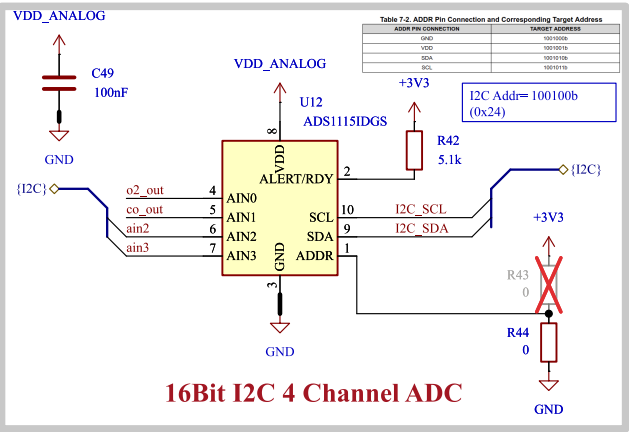
Speaker power switch

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Sheet Title: Power Supply	File Name: PowerSupply.kicad_sch	Designer: Oriol Reverte	Date:	Revision:
Sheet Path: /Project Architecture/Power Supply/	Reviewer:	Size: A3	Sheet: 11 of 12	

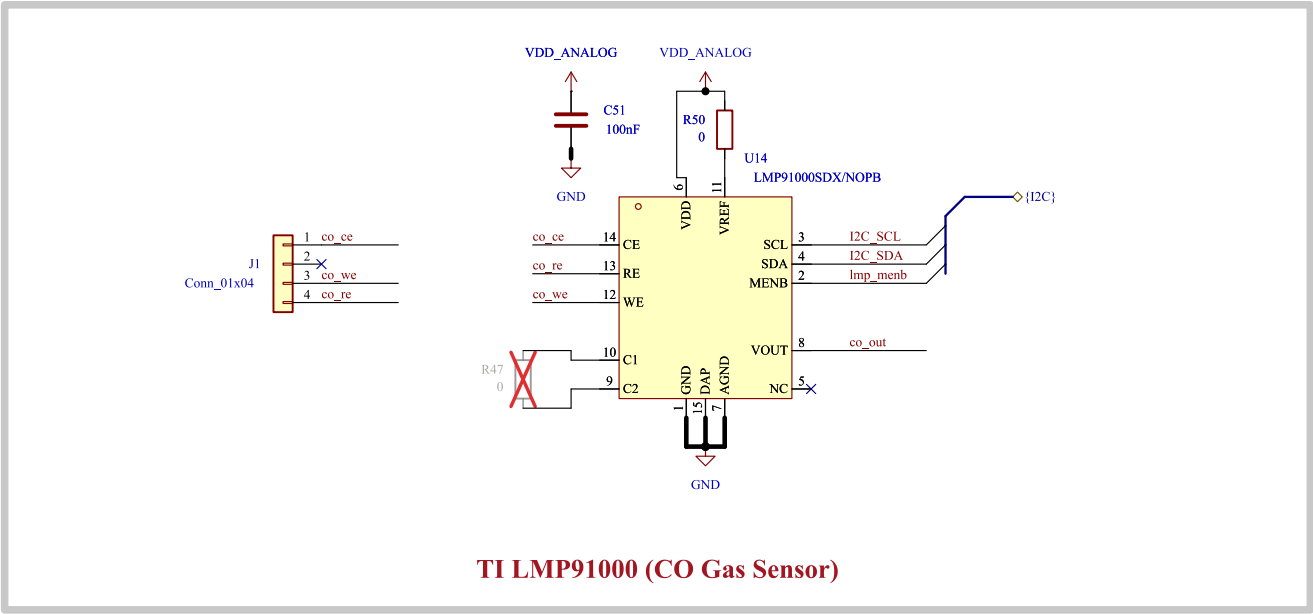
[12] Gas Sensors



O2 Gas sensor Op Amp Interface



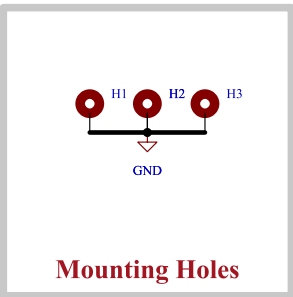
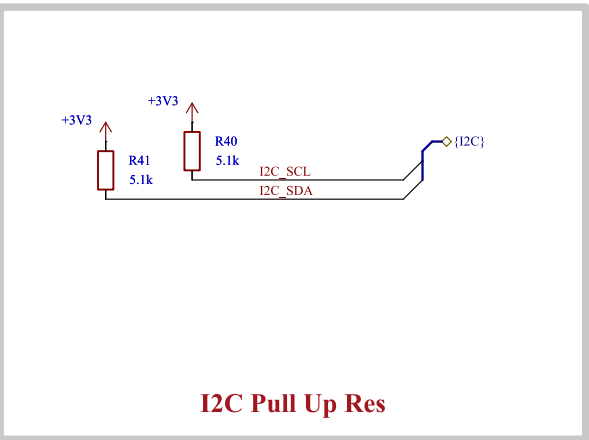
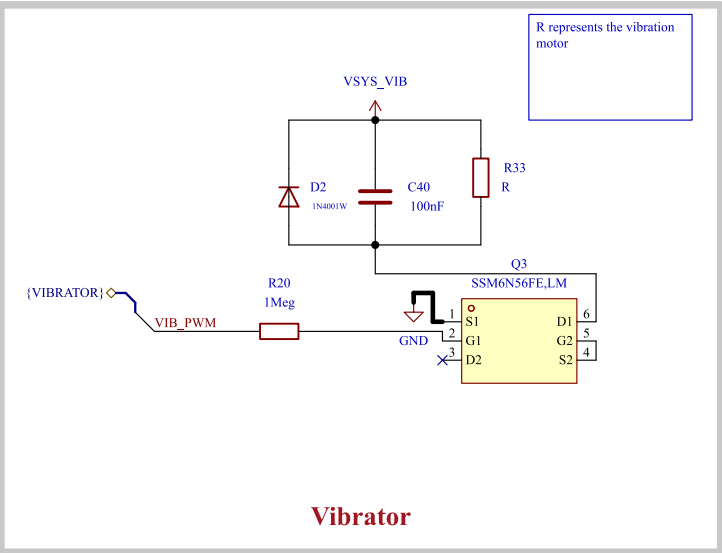
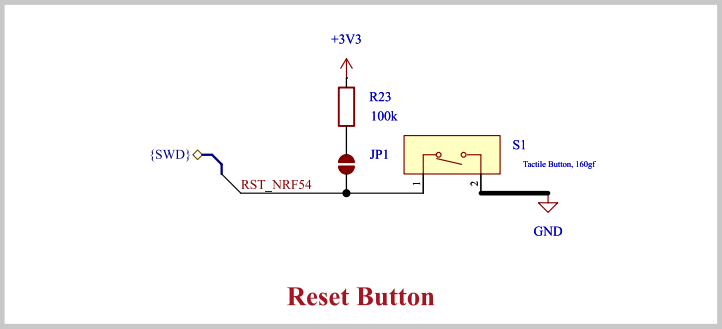
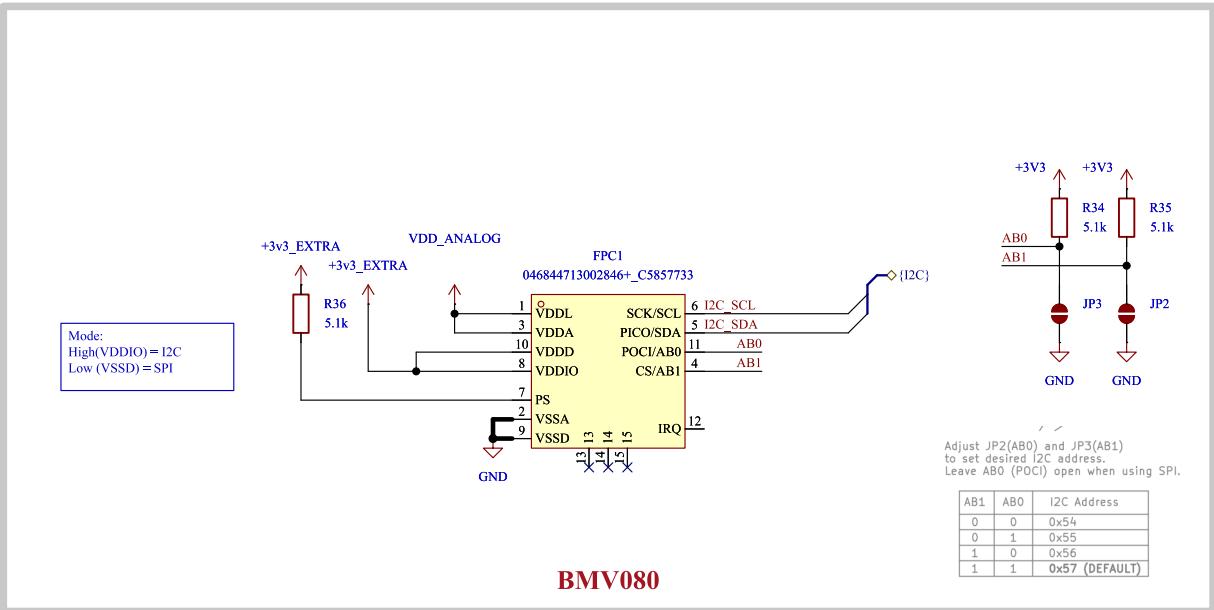
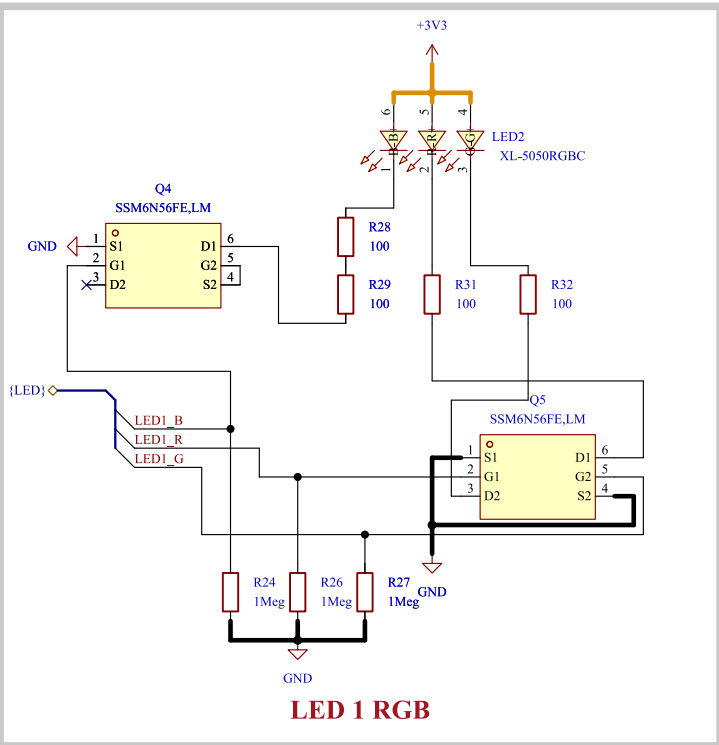
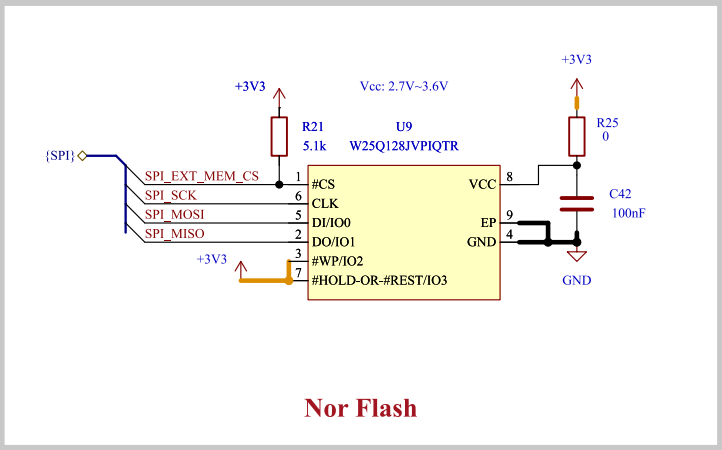
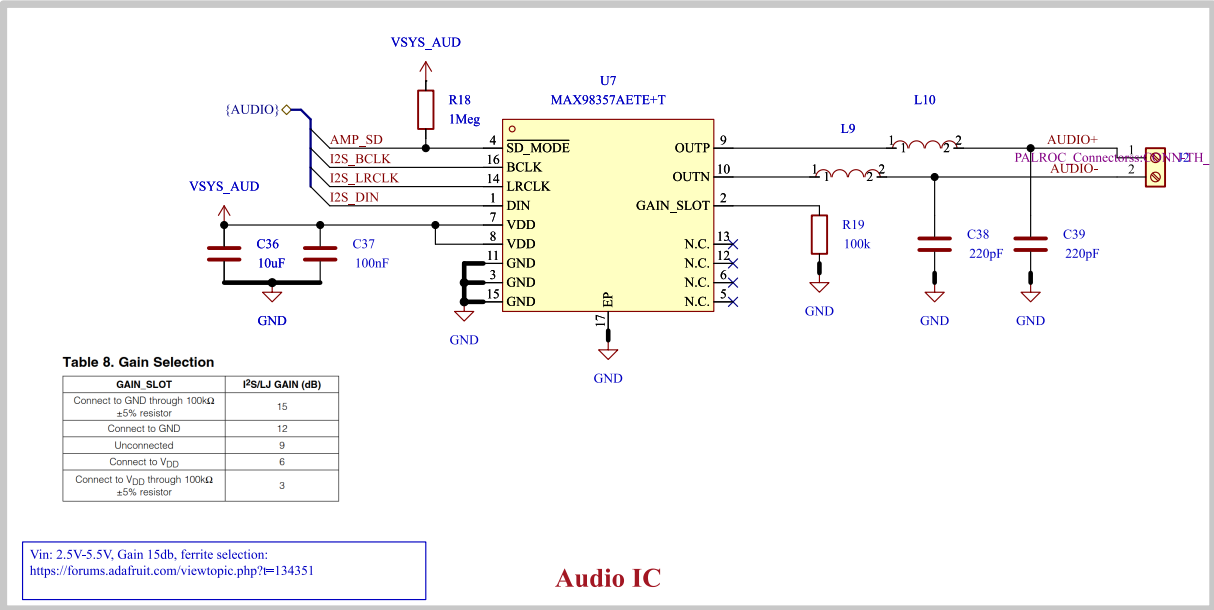
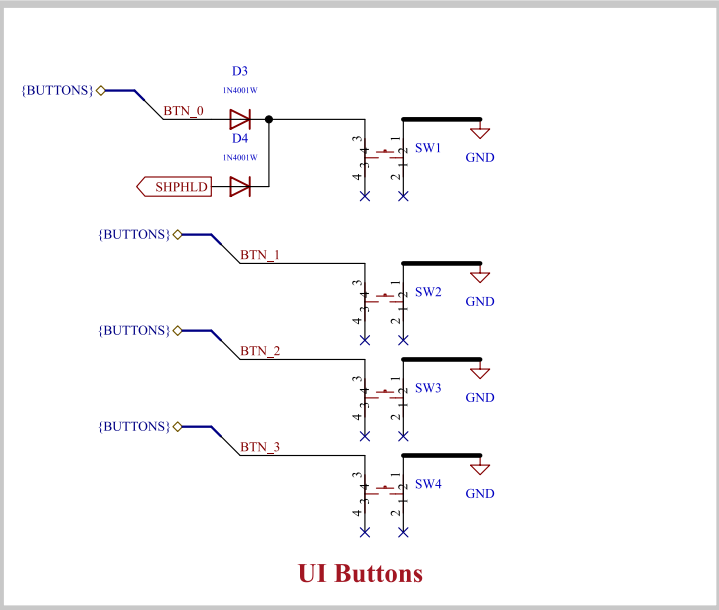
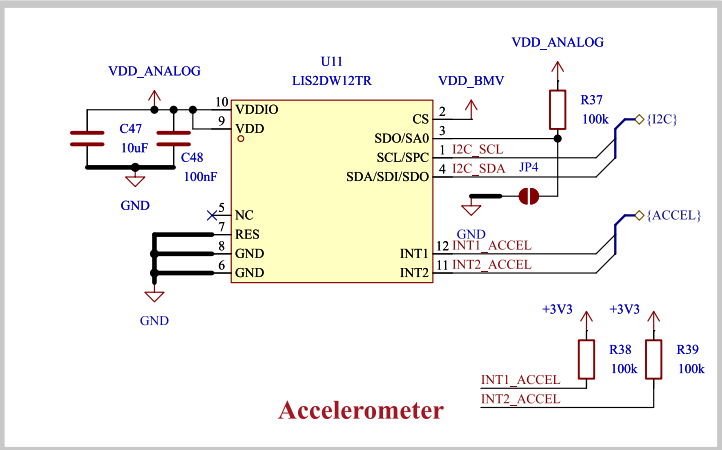
16Bit I2C 4 Channel ADC



TI LMP91000 (CO Gas Sensor)

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	Gas Sensors	Gas Sensors.kicad_sch	Oriol Reverte		
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	/Project Architecture/Gas Sensors/			A3	12 of 12

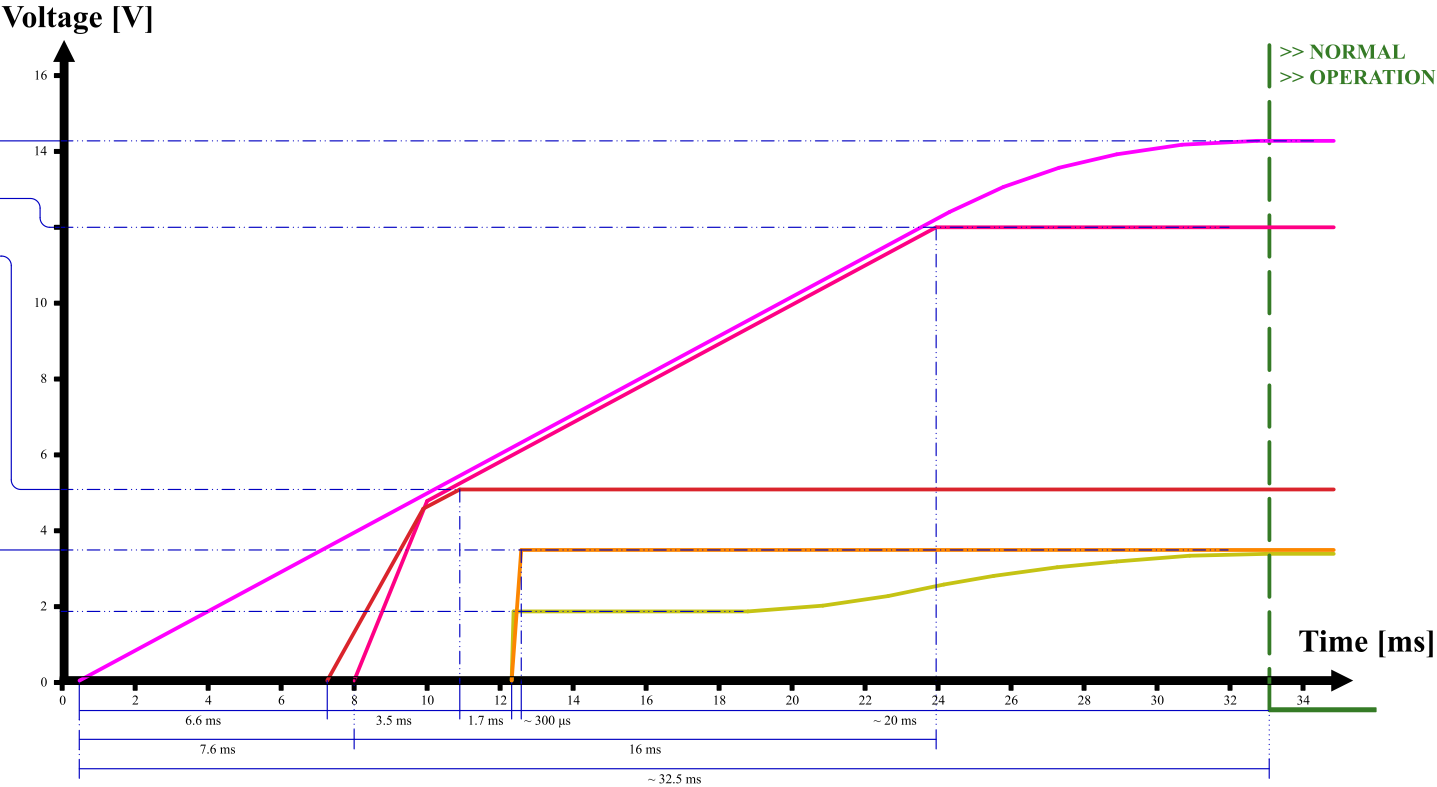
[14] Misc



Sheet Title: Misc	Comments:		Company:		Variant: PRELIMINARY	Git Hash: 4dfdc44
	Sheet Path: /Project Architecture/Misc/		Board Name: aeris_pcb		Project Name: Palroc One	
	Designer: Oriol Reverte		Date:		Revision: 14 of 12	

[6] Power - Sequencing

NAME	SOURCE	LEVEL
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%



	Comments:	Company: Palroc Labs		Variant: PRELIMINARY	Git Hash: 4dfdc44
		Board Name: aeris_pcb		Project Name: Palroc One	
	Sheet Title: Power - Sequencing	File Name: Power - Sequencing.kicad_sch	Designer: Oriol Reverte	Date: 2025-01-12	Revision: + (Unreleased)
	Sheet Path: /Power - Sequencing/		Reviewer:	Size: A4	Sheet: 6 of 12

[7] Revision History

	Comments:	Company: Palroc Labs		Variant: PRELIMINARY	Git Hash: 4dfdc44
		Board Name: aeris_pcb		Project Name: Palroc One	
	Sheet Title: Revision History	File Name: Revision History.kicad_sch	Designer: Oriol Reverte	Date: 2025-01-12	Revision: + (Unreleased)
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